

Los Sabrossos

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1 STRINGS

1.1 Hashing

```
1 struct Hash {
2     long long P=1777771,MOD[2],PI[2];
3     vector<long long> h[2],pi[2];
4     Hash(string& s){
5         MOD[0]=999727999;MOD[1]=1070777777;
6         PI[0]=325255434;PI[1]=10018302;
7         for(int k = 0; k < 2; k++) h[k].resize(s.size()+1),pi[k].resize(s.size()+1);
8         for(int k = 0; k < 2; k++){
9             h[k][0]=0;pi[k][0]=1;
10            long long p=1;
11            for(int i = 1; i < s.size()+1; i++){
12                h[k][i]=(h[k][i-1]+p*s[i-1])%MOD[k];
13                pi[k][i]=(1LL*pi[k][i-1]*PI[k])%MOD[k];
14                p=(p*P)%MOD[k];
15            }
16        }
17    }
18    long long get(int s, int e){
19        long long h0=(h[0][e]-h[0][s]+MOD[0])%MOD[0];
20        h0=(1LL*h0*pi[0][s])%MOD[0];
21        long long h1=(h[1][e]-h[1][s]+MOD[1])%MOD[1];
22        h1=(1LL*h1*pi[1][s])%MOD[1];
23        return (h0<<32)|h1;
24    }
25};
```

1.2 Kmp

```
1 vector<int> kmp(string s){
2     int n = s.size();
3     vector<int> pi(n);
4     for(int i = 1; i < n; i++){
5         int j = pi[i-1];
6         while(j > 0 and s[i] != s[j]){
7             j = pi[j-1];
8         }
9         if(s[i] == s[j]){
10            j++;
11        }
12        pi[i] = j;
13    }
14    return pi;
15}
16 //Pattern Matching
17 vector<int> ocurrencia(string texto, string patron) {
18     vector<int> P = kmp(patron);
19     int k = 0;
20     vector<int> M;
21     repl(i,0,texto.size()) {
22         while (k > 0 and patron[k] != texto[i]) k = P[k - 1];
23         if (patron[k] == texto[i]) k++;
24         if (k == patron.size()) {
25             M.pb(i - k + 1);
26             k = P[k - 1];
27         }
28     }
29     return M;
```

```

30 }
31
32 //Automata
33 void genAut(string &s) {
34     int n = s.size();
35
36     vector<vector<int>> aut(n+1, vector<int>(26));
37     vector<int> pi = kmp(s);
38
39     for(int i = 0 ; i < n; i++){
40         aut[i][s[i]-'a'] = i+1;
41     }
42
43     for(int i = 0; i < n+1; i++){
44         for(int c = 0; c < 26; c++){
45             if (aut[i][c]) continue;
46             if (i < n and s[i] == 'a' + c) aut[i][c] = i+1;
47             else if (i > 0) aut[i][c] = aut[pi[i-1]][c];
48             else aut[i][c] = 0;
49         }
50     }
51 }

```

2 TREE ALGORITHMS

2.1 Binary lifting

```

1  const int N = 1e5;
2  const int LOG = 20;
3  int up[N][LOG];
4  vector<int> G[N];
5  //Find parents
6  void dfs(int u){
7      for(int v : G[u]){
8          up[v][0] = u;
9          for(int j = 1; j < LOG; j++){
10             up[v][j] = up[up[v][j-1]][j-1];
11         }
12         dfs(v);
13     }
14 }
15 int get_K_parent(int a, int k){
16     for(int j = LOG-1; j >= 0; j++){
17         if(k&(1<<j)){
18             a = up[a][j];
19         }
20     }
21     return a;
22 }

```

2.2 Lowest common ancestor

```

1  const int N = 1e5;
2  const int LOG = 20;
3  vector<int> G[N];
4  int depth[N];
5  int up[N][LOG];
6  // Assign levels and get parents
7  void dfs(int u){

```

```

8     for(int v : G[u]){
9         depth[v] = depth[u] + 1;
10        up[v][0] = u;
11        for(int j = 1; j < LOG; j++){
12            up[v][j] = up[up[v][j-1]][j-1];
13        }
14        dfs(v);
15    }
16 }
17 int get_lca(int a, int b){
18     if(depth[a] < depth[b]) swap(a,b);
19     int k = depth[a] - depth[b];
20     for(int j = LOG-1; j >= 0; j--){
21         if(k & (1<<j)){
22             a = up[a][j];
23         }
24     }
25     if(a == b) return a;
26     for(int j = LOG-1; j >= 0; j--){
27         if(up[a][j] != up[b][j]){
28             a = up[a][j];
29             b = up[b][j];
30         }
31     }
32     return up[a][0];
33 }

```

3 DATA STRUCTURES

3.1 Disjoint set union

```

1  const int N = 1e5;
2  struct DSU{
3      int num;
4      vector<int> parent;
5      vector<int> sizes;
6      DSU(){};
7      DSU(int n){
8          init(n);
9      }
10     void init(int n){
11         num = n;
12         parent.resize(n+1);
13         sizes.resize(n+1);
14         for(int i = 1; i <= n ; i++){
15             parent[i] = i;
16             sizes[i] = 1;
17         }
18     }
19     int find(int a){
20         if(a == parent[a]) return a;
21         return parent[a] = find(parent[a]);
22     }
23     void join(int a, int b){
24         int x = find(a);
25         int y = find(b);
26         if(x == y) return;
27         num--;
28         if(sizes[x] < sizes[y]){
29             parent[x] = y;
30             sizes[y] += sizes[x];

```

```

31     }
32     else{
33         parent[y] = x;
34         sizes[x]+=sizes[y];
35     }
36
37 }
38 };

```

3.2 Segment tree

```

1  const int N = 1e5;
2  struct info{
3      int num;
4  };
5  info ST[4*N];
6  info ope(info u, info v){
7      return {u.num+v.num};
8  }
9  void build(int in, int L, int R){
10     if(L == R){
11         cin >> ST[in].num;
12     }
13     else{
14         int mid = (L+R)>>1;
15         build((2*in),L,mid);
16         build((2*in)+1,mid+1,R);
17         ST[in] = ope(ST[(2*in)], ST[(2*in)+1]);
18     }
19 }
20 void update(int in, int L, int R, int pos, int val){
21     if(L == R){
22         ST[in].num = val;
23     }
24     else{
25         int mid = (L+R)>>1;
26         if(pos <= mid){
27             update(2*in,L,mid,pos,val);
28         }
29         else{
30             update((2*in)+1,mid+1,R,pos,val);
31         }
32         ST[in] = ope(ST[2*in], ST[(2*in)+1]);
33     }
34 }
35 info get(int in, int L, int R, int l, int r){
36     if(r < L or R < l) return {0};
37     if(l <= L and R <= r) return ST[in];
38     int mid = (L+R)>>1;
39     info left = get(2*in,L,mid,l,r);
40     info right = get((2*in)+1,mid+1,R,l,r);
41     return ope(left,right);
42 }
43 // v = {1 2 3 4 5}
44 // get(1, 0, n-1, 1, 3) = 2 + 3 + 4 = 9

```

3.3 Sparse table

```

1  const int N = 1e5;
2  const int LOG = 20;

```

```

3 int v[N]; //values
4 int st[N][LOG]; //Sparse Table
5 int lg[N]; // log values
6 int n;
7 void build(){
8     for(int i = 0; i < n; i++) st[i][0] = v[i];
9
10    for(int i = 1; i < LOG; i++){
11        for(int j = 0; j < n - (1<<i) + 1; j++){
12            st[j][i] = min(st[j][i-1], st[j+(1<<(i-1))][i-1]);
13        }
14    }
15    lg[1] = 0;
16    for(int i = 2; i <= n; i++) lg[i] = lg[i/2] + 1;
17 }
18 int query(int L, int R){
19     int k = lg[R-L+1];
20     return min(st[L][k], st[R-(1<<k)+1][k]);
21 }
22 // v = [5, 2, 4, 7, 1, 3]
23 // query(1, 4) → [2, 4, 7, 1]

```

4 MATH

4.1 Binary exponentiation

```

1 const long long mod = 998244353;
2 long long pote(long long a, long long b){
3     long long ans = 1ll;
4     while(b){
5         if(b&1) ans = (ans * a)%mod;
6         b>>=1;
7         a = (a * a)%mod;
8     }
9     return ans;
10 }

```

4.2 Modular inverse

```

1 const long long N = 1e6 + 10;
2 const int MOD = 1e9+7;
3 long long fact[N];
4 long long invfact[N];
5 void init(){
6     fact[0] = 1;
7     for(long long i = 1; i < N; i++)
8         fact[i] = fact[i-1] * i % MOD;
9
10    invfact[N-1] = pote(fact[N-1], MOD-2);
11
12    for(long long i = N-2; i >= 0; i--)
13        invfact[i] = invfact[i+1] * (i+1) % MOD;
14 }
15
16 long long nCk(long long n, long long k){
17     if(k < 0 || k > n) return 0;
18     return fact[n] * invfact[k] % MOD * invfact[n-k] % MOD;
19 }

```


4.3 Pollard rho rabin miller

```
1 map<long long, long long> fact;
2
3 long long mcd(long long a, long long b) {
4     a = abs(a);
5     b = abs(b);
6     while (a != 0) {
7         long long r = b % a;
8         b = a;
9         a = r;
10    }
11    return b;
12 }
13 //Rabin Miller
14 bool primo(long long n) {
15     if (n < 2) return false;
16     if (n == 2) return true;
17     long long D = n - 1, S = 0;
18     while (D % 2 == 0) {
19         D /= 2;
20         S++;
21     }
22     static const long long STEPS = 1611;
23     for(long long pasos = 0; pasos < STEPS; pasos++){
24         const long long A = 1 + rand() % (n - 1);
25         long long M = pote(A, D, n);
26         if (M == 1 || M == (n - 1)) goto next;
27         for(long long k = 0; k < S-1; k++){
28             M = ((M) * M) % n;
29             if (M == (n - 1)) goto next;
30         }
31         return false;
32     next;;
33     }
34     return true;
35 }
36
37 //Rho de pollard
38 long long factor(long long n) {
39     static long long A, B;
40     A = 1 + rand() % (n - 1);
41     B = 1 + rand() % (n - 1);
42     #define fun(x) (((x) * (x + B)) % n + A)
43
44     long long x = 2, y = 2, d = 1;
45     while (d == 1 || d == -1) {
46         x = fun(x);
47         y = fun(fun(y));
48         d = mcd(x - y, n);
49     }
50     return abs(d);
51 }
52
53 void factorize(long long n){
54     assert(n > 0);
55     while (n > 1 && !primo(n)) {
56         long long fx;
57         do {
58             fx = factor(n);
59         } while (fx == n);
60
61         n /= fx;
```

```

62     factorize(fx);
63
64     for (auto &it : fact) {
65         while (n % it.first == 0) {
66             n /= it.first;
67             it.second++;
68         }
69     }
70 }
71 if (n > 1) fact[n]++;
72 }

```

4.4 Sieve of eratosthenes

```

1  const int N = 1e7;
2  int sieve[N];
3  void make(){
4      for(int i = 0; i < N; i++) sieve[i] = 1;
5      sieve[0] = sieve[1] = 0;
6      for (int i = 2; i < N; i++)
7      {
8          if(sieve[i]){
9              for (int j = i*i; j < N; j+=i)
10                 {
11                     sieve[j] = 0;
12                 }
13             }
14         }
15     }

```

5 GRAPH ALGORITHMS

5.1 Bfs

```

1  const int N = 1e5;
2
3  vector<int> G[N];
4  int vis[N];
5  void bfs(int source){
6      queue<int> q;
7      q.push(source);
8      vis[source] = 0;
9      while(!q.empty()){
10         int u = q.front();
11         q.pop();
12         for(int v : G[u]){
13             if(vis[v] == -1){
14                 vis[v] = vis[u] + 1;
15                 q.push(v);
16             }
17         }
18     }
19 }

```

5.2 Dfs

```

1  const int N = 1e5;
2
3  vector<int> G[N];
4  int vis[N];
5
6  void dfs(int u){
7      vis[u] = 1;
8      for(int v : G[u]){
9          if(vis[v] == -1){
10             dfs(v);
11         }
12     }
13 }

```

5.3 Dijkstra

```

1  const int N = 1e5;
2
3  int vis[N];
4  vector<pair<int,int>> G[N];
5
6  void dijkstra(int x){
7      priority_queue<pair<int,int>, vector<pair<int,int>>, greater<pair<int,int>>> pq;
8      pq.push({0,x});
9      vis[x] = 0;
10     while(!pq.empty()){
11         auto y = pq.top();
12         pq.pop();
13         int u = y.second;
14         int cost = y.first;
15         for(auto v : G[u]){
16             int cur = cost + v.second;
17             if(cur < vis[v.first]){
18                 vis[v.first] = cur;
19                 pq.push({cur,v.first});
20             }
21         }
22     }
23 }

```

5.4 Kruskal mst

```

1  const int N = 1e5;
2  struct DSU{
3      int num;
4      vector<int> parent;
5      vector<int> sizes;
6      DSU(){};
7      DSU(int n){
8          init(n);
9      }
10     void init(int n){
11         num = n;
12         parent.resize(n+1);
13         sizes.resize(n+1);
14         for(int i = 1; i <= n ; i++){
15             parent[i] = i;
16             sizes[i] = 1;
17         }
18     }
19 }

```

```

18     }
19     int find(int a){
20         if(a == parent[a]) return a;
21         return parent[a] = find(parent[a]);
22     }
23     void join(int a, int b){
24         int x = find(a);
25         int y = find(b);
26         if(x == y) return;
27         num--;
28         if(sizes[x] < sizes[y]){
29             parent[x] = y;
30             sizes[y]+=sizes[x];
31         }
32         else{
33             parent[y] = x;
34             sizes[x]+=sizes[y];
35         }
36     }
37 };
38 int n;
39 vector<pair<int,pair<int,int>>> List;
40 int Kruskal_MST(){
41     sort(List.begin(),List.end());
42     DSU dsu(n);
43     int mst = 0, t = 1;
44     for(int i = 0; i < List.size(); i++){
45         int w = List[i].first; //weight
46         int a = List[i].second.first; //at
47         int b = List[i].second.second; //to
48         if(dsu.find(a) != dsu.find(b)){
49             dsu.join(a,b);
50             mst+=w;
51             t++;
52         }
53         if(t == n) break;
54     }
55     return mst;
56 }

```

5.5 Topological sort kahn algorithm

```

1  const int N = 1e5;
2
3  int indegree[N];
4  vector<int> G[N];
5  int n;
6  vector<int> Kahn(){
7      queue<int> q;
8      vector<int> ans;
9      for(int i = 0; i < n; i++){
10         if(indegree[i] == 0) q.push(i);
11     }
12     while (!q.empty()){
13         int v = q.front();
14         q.pop();
15         ans.push_back(v);
16         for(int u : G[v]){
17             indegree[u]--;
18             if(indegree[u] == 0){
19                 q.push(u);
20             }

```

```
21     }
22 }
23 //if(ans.size() != n) grafo no DAG
24 return ans;
25 }
```